

ABHINAV MARLINGAPLAR

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LinkedIn



GitHub



Portfolio



LeetCode

Education

Indian Institute of Information Technology, Kottayam

Aug 2023 – Present

B.Tech in Computer Science (Artificial Intelligence & Data Science)

CGPA: 8.24 / 10

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Cloud Computing, Machine Learning

Experience

Agentic AI Intern

May 2026 – Present

Koorier Inc.

- Engineered a production-grade multi-agent AI system using LangGraph to automatically detect and correct malformed shipping addresses for a Canadian last-mile logistics platform.
- Designed a 4-agent workflow (Researcher, Validator, Critic, Communicator) integrating StreetPerfect SERP, Canada Post AddressComplete, and HERE Geocoding APIs to resolve address defects with automated reflection and confidence scoring.
- Built and optimized LLM-powered agents with deterministic validation logic, reducing inference cost and improving system reliability across a range of real-world address edge cases.
- Developed production tooling including audit logging, structured workflow state management, and a Playwright-based headless scraper for Canada Post AddressComplete with anti-detection handling.

Projects

Ops Pilot – Cloud-Native CI/CD Automation Platform

May 2026

Tools: React.js, Node.js, RabbitMQ, Docker, Kubernetes, PostgreSQL (NeonDB), Socket.IO

- Developed a cloud-native CI/CD automation platform inspired by Jenkins and GitHub Actions using React, Node.js, RabbitMQ, Docker, Kubernetes, and PostgreSQL (NeonDB).
- Built asynchronous pipeline execution with distributed worker pods for automated build, test, Docker image generation, and Kubernetes deployment workflows.
- Implemented GitHub webhook integration, real-time build monitoring using Socket.IO, and fault-tolerant job processing with RabbitMQ retry mechanisms.
- **Live:** ops-pilot-rho.vercel.app

NLP Studio – AI-Powered NLP Platform

Dec 2025

Tools: MERN Stack, Supabase pgvector, Groq API, Redis, JWT, Vercel, Render

- Built a full-stack NLP platform with a RAG document chatbot (multi-turn memory), a 3-variant paraphrasing engine, and a text analytics service returning sentiment, urgency scores, and NPS estimates.
- Engineered a RAG pipeline using Xenova Transformers (MiniLM L6 v2), Supabase pgvector similarity search, and Groq LLM with per-session conversation history in Redis enabling context-aware dialogue.
- Implemented dual-layer Redis caching (embeddings 24h, responses 1h), full userId-scoped session isolation across Supabase and localStorage, and JWT auth with HTTP-only session cookies and silent Axios token refresh.
- **Live:** nlp-studio.vercel.app

House Price Prediction

June 2025

Tools: Python, Pandas, Scikit-learn, XGBoost

- Built a regression pipeline using XGBoost with hyperparameter tuning, achieving $R^2 = 0.9204$ on test data.
- Applied preprocessing techniques including feature engineering, ordinal encoding, and cross-validation for improved generalization.
- Validated performance through Kaggle submission with an RMSE score of 0.13469.

Technical Skills

- **Languages:** C++, Python, JavaScript, SQL
- **Web Technologies:** React.js, HTML5, CSS3, Node.js, Express.js, REST APIs
- **Databases:** MongoDB, PostgreSQL
- **DevOps & Infrastructure:** Docker, Kubernetes, RabbitMQ
- **Tools:** Git, GitHub, Postman

Additional Highlights

- Solved 450+ DSA problems on LeetCode spanning graphs, dynamic programming, trees, and sliding window techniques.
- Built and deployed multiple full-stack and AI-powered applications with experience in system design, API integration, and production deployment.